

Strong Orientation

A **strong orientation** of an undirected graph is an assignment of a direction to each edge that makes it a **strongly connected graph**. That is, after the *orientation* we should be able to visit any vertex from any vertex by following the directed edges.

Solution

Of course, this cannot be done to every graph. Consider a **bridge** in a graph. We have to assign a direction to it and by doing so we make this bridge "crossable" in only one direction. That means we can't go from one of the bridge's ends to the other, so we can't make the graph strongly connected.

Now consider a **DFS** through a bridgeless connected graph. Clearly, we will visit each vertex. And since there are no bridges, we can remove any DFS tree edge and still be able to go from below the edge to above the edge by using a path that contains at least one back edge. From this follows that from any vertex we can go to the root of the DFS tree. Also, from the root of the DFS tree we can visit any vertex we choose. We found a strong orientation!

In other words, to strongly orient a bridgeless connected graph, run a DFS on it and let the DFS tree edges point away from the DFS root and all other edges from the descendant to the ancestor in the DFS tree.

The result that bridgeless connected graphs are exactly the graphs that have strong orientations is called **Robbins' theorem**.

Problem extension

Let's consider the problem of finding a graph orientation so that the number of SCCs is minimal.

Of course, each graph component can be considered separately. Now, since only bridgeless graphs are strongly orientable, let's remove all bridges temporarily. We end up with some number of bridgeless components (exactly *how many components there were at the beginning + how many bridges there were*) and we know that we can strongly orient each of them.